

When Correlation Means Causation

Pragmatic factors modulate causal implicatures in decision-making contexts

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Correlation is not causation. But...

Reichenbach gives us a rationalization for causal inference

Stats 101

correlation $\neq$ causation



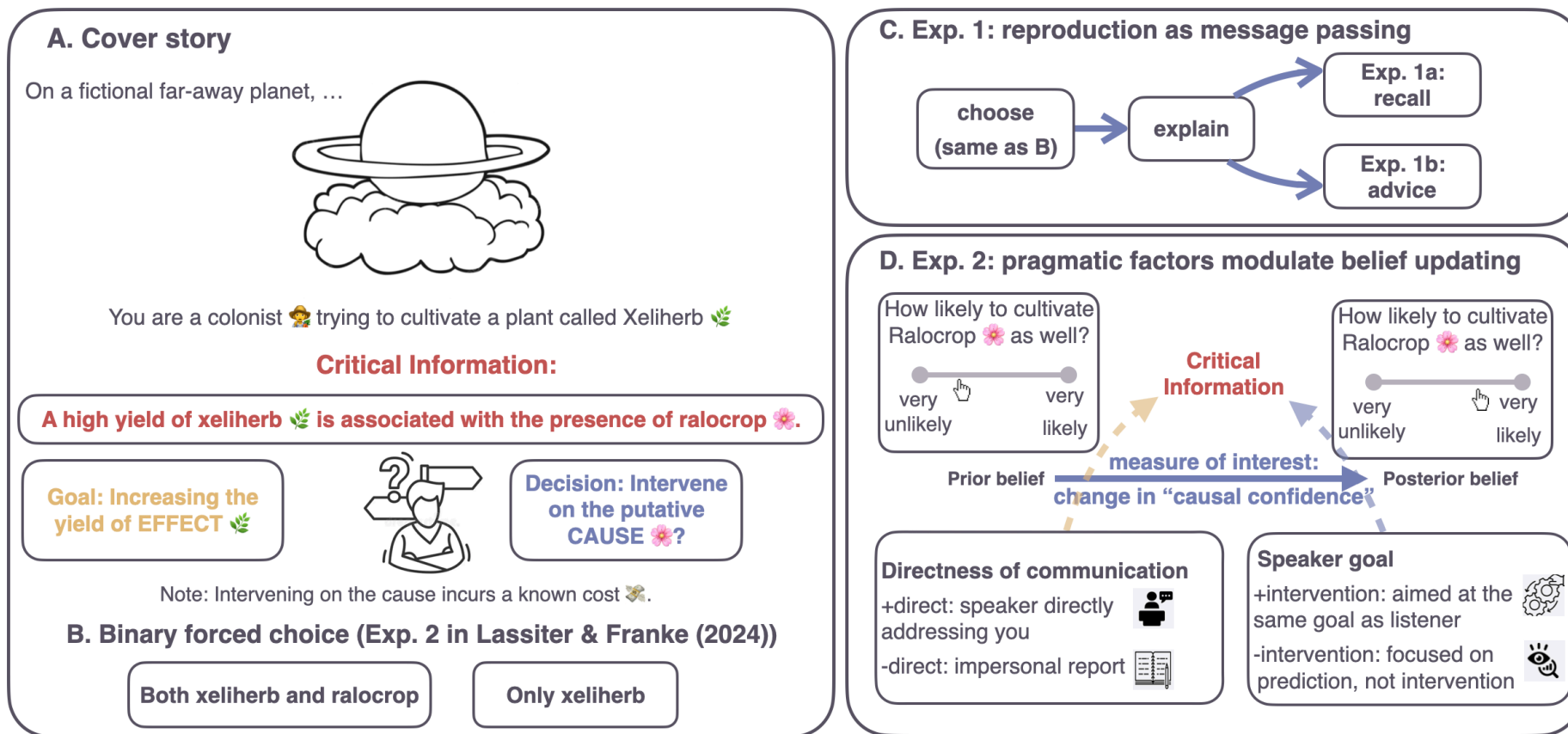
Reichenbach

**every non-accidental
correlation is due to some
causal structure**

In communication, it can therefore be rational to infer a speaker's causal belief from "A is associated with B" — even though causation is not part of the literal meaning.

Overview

Does this inference affect decisions, and does communicative context matter?



Action
cultivate both plants?



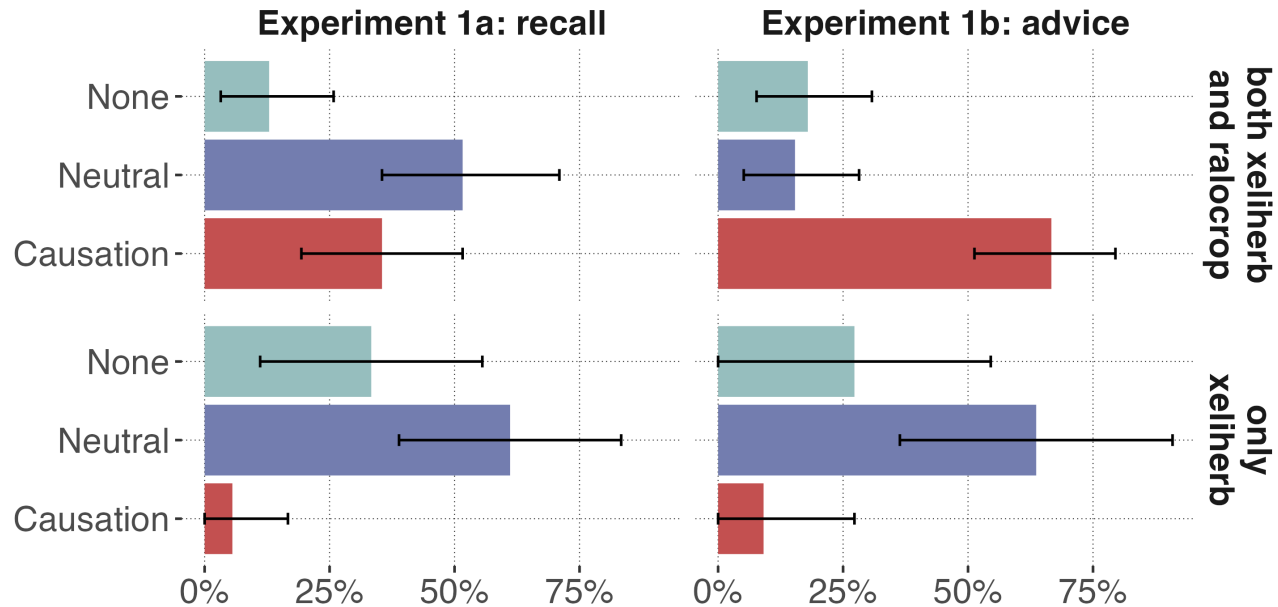
Language
recall or advise



Belief
posterior – prior

Experiment 1: Reproduction as message passing

Free production after the same intervention choice · N = 99



Those who chose **both plants** used causal language more often.

The pattern appeared in recall and was clearest when participants advised another outpost.

Recall P = .98

Advice P = .99

Experiment 2: Pragmatic factors modulate belief updating

N = 404

Directness

Direct: selected for you

Indirect: found in an old note

Speaker goal

Cultivation: improve yield

Localization: find wild xeliherb

Prior judgment
willingness to cultivate
ralocrop



Posterior judgment
willingness to cultivate
ralocrop

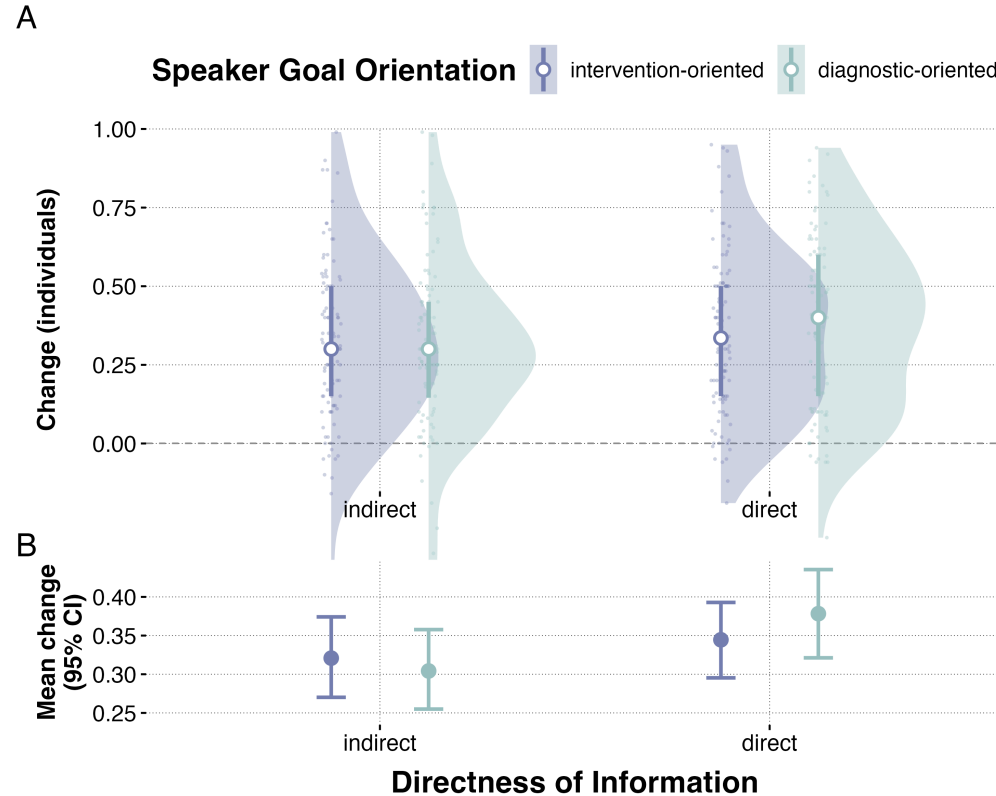
correlational information

Primary dependent measure

posterior – prior

within-participant causal belief change

Correlation increased causal commitment



89–91% became more willing to cultivate ralocrop.

Overall +.31 [.28, .34]

Directness +.06 [.02, .09]

Interaction +.07 · P = .97

Direct reports led to larger updates. The speaker-goal contrast appeared mainly when the report was addressed to the listener.

Take-home

The same correlational statement produced different degrees of causal commitment across communicative contexts.

This pattern is difficult to explain as a context-insensitive bias. It supports a pragmatic account: listeners reason about why a particular speaker chose a correlational formulation in a particular setting.

Thank you

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